Disease Surveillance System for Nigeria

Algorithm Document: Case Percentages, Disease Growth and Outbreak Alerts

Platform	Windows desktop application (PowerShell / Windows Forms, packaged as .exe)
Coverage	Nigeria (state level); extendable to other countries later
Intended users	Public health workers, doctors, surgeons and health agencies such as WHO
Prepared by	_____ Date: _____

1. Project Title and Purpose

Disease Surveillance System for Nigeria. The application reads reported disease case records for Nigerian states and shows which diseases are most prominent in each state, what percentage of a state's reported cases each disease accounts for, how fast each disease is growing, and which diseases should raise an outbreak alert. This document describes the algorithm behind those results.

2. Objectives

- Calculate, for every state, the percentage share of each disease in that state's reported cases.
- Measure the growth (trend) of each disease by comparing the latest period with the previous period.
- Classify every disease record as OUTBREAK, WATCH or NORMAL using clear, adjustable rules.
- Present the results as a table, charts and an alert list so health workers can act quickly.
- Keep the method simple enough to implement in PowerShell and to explain in an assignment.

3. Inputs

The algorithm takes a list of records (loaded from a CSV file, or the built-in sample data). Each record describes one disease in one state for two consecutive periods.

Field	Type	Meaning	Example
State	Text	Nigerian state (or FCT)	Borno
Zone	Text	Geopolitical zone of the state	North East
Disease	Text	Disease name	Cholera
Cases	Integer	Reported cases in the latest period	1500
PrevCases	Integer	Reported cases in the previous period	600

User selections: State (or All states), Disease (or All diseases). Adjustable thresholds: OUTBREAK_GROWTH = 50, OUTBREAK_SHARE = 10, WATCH_GROWTH = 25.

The records supplied with the project are **sample data for demonstration only**; they are not official figures. Real data should come from an official source such as the Nigeria Centre for Disease Control (NCDC).

4. Processing Steps

Step 1. Load the records and convert Cases and PrevCases to numbers.

Step 2. Group the records by State.

Step 3. For each state, add up Cases of all its diseases to get Total(State).

Step 4. For each record, calculate Percent (the disease's share of its state's cases).

Step 5. For each record, calculate Growth (change since the previous period).

Step 6. Apply the decision rules to set Alert to OUTBREAK, WATCH or NORMAL.

Step 7. Filter the results by the State and Disease chosen by the user.

Step 8. Rank diseases in the selection, find the most prominent disease, total the cases and count the OUTBREAK alerts.

Step 9. Display the statistics table, share chart, trend chart and alert list; allow export to CSV.

5. Calculations

Quantity	Formula	Notes
Total(State)	sum of Cases for all records of that state	Denominator for percentages
Percent	(Cases / Total(State)) x 100	Rounded to 1 decimal place; 0 if Total = 0
Growth (%)	((Cases - PrevCases) / PrevCases) x 100	Rounded to 1 decimal place; 0 if PrevCases = 0
Most prominent disease	disease with the highest Percent (or highest summed Cases when many states are selected)	Shown on the dashboard

Worked example: Borno (Total = 5450 cases)

Disease	Cases	Prev	Percent calculation	Percent	Growth	Alert
Malaria	3000	2900	3000 / 5450 x 100	55.0%	+3.4%	NORMAL
Cholera	1500	600	1500 / 5450 x 100	27.5%	+150.0%	OUTBREAK
Measles	950	800	950 / 5450 x 100	17.4%	+18.8%	NORMAL

Cholera: Growth = (1500 - 600) / 600 x 100 = 150%, and its share is 27.5%, which is at least 10%, so the OUTBREAK rule is met.

6. Decision Rules

Order	Condition	Alert	Meaning
1	Growth >= 50% **and** Percent >= 10%	OUTBREAK	Fast growth in a disease that makes up a significant share of cases
2	Otherwise, Growth >= 25%	WATCH	Noticeable growth; monitor closely
3	Otherwise	NORMAL	No unusual increase

Rules are checked in order and the first one that is true is used. The share condition in rule 1 prevents a very rare disease with a small absolute rise from being flagged as an outbreak (it becomes WATCH instead). The threshold values (50, 10, 25) are placeholders and should be confirmed by public health experts.

7. Pseudocode

```
ALGORITHM DiseaseSurveillance
INPUT : records[] (State, Zone, Disease, Cases, PrevCases)
        selectedState, selectedDisease
OUTPUT: results[], topDisease, totalCases, outbreakCount

CONSTANTS OUTBREAK_GROWTH = 50, OUTBREAK_SHARE = 10, WATCH_GROWTH = 25

BEGIN
  FOR EACH state IN distinct(records.State)
      total[state] = SUM(Cases) of records where State = state
  END FOR

  results = empty list
  FOR EACH r IN records
      IF total[r.State] > 0 THEN
          percent = ROUND(r.Cases / total[r.State] * 100, 1)
      ELSE
          percent = 0
      END IF

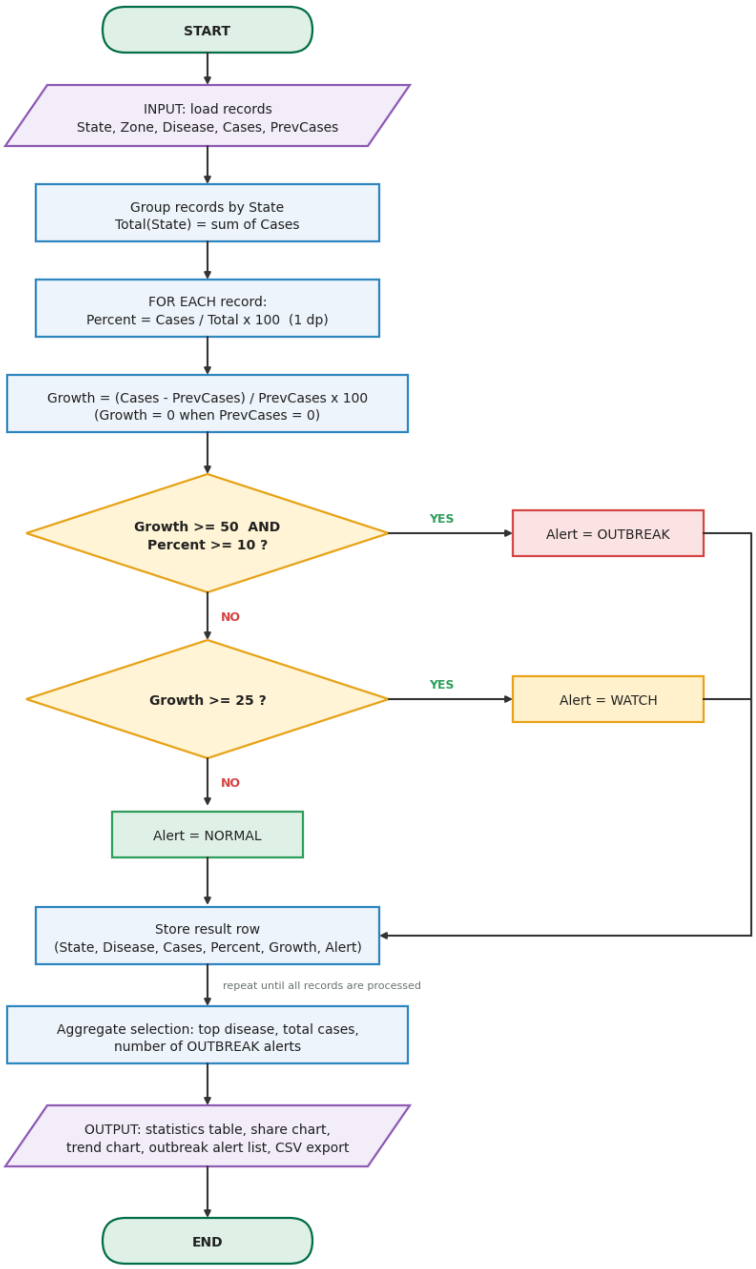
      IF r.PrevCases > 0 THEN
          growth = ROUND((r.Cases - r.PrevCases) / r.PrevCases * 100, 1)
      ELSE
          growth = 0
      END IF

      IF growth >= OUTBREAK_GROWTH AND percent >= OUTBREAK_SHARE THEN
          alert = "OUTBREAK"
      ELSE IF growth >= WATCH_GROWTH THEN
          alert = "WATCH"
      ELSE
          alert = "NORMAL"
      END IF

      ADD (r.State, r.Zone, r.Disease, r.Cases, percent, growth, alert) TO results
  END FOR

  selection     = results FILTERED BY selectedState and selectedDisease
  topDisease    = disease with the largest summed Cases in selection
  totalCases    = SUM(Cases) in selection
  outbreakCount = COUNT of rows in selection where alert = "OUTBREAK"
  DISPLAY selection (table), chart of percentages, trend chart, alert list
END
```

8. Flowchart


START
INPUT: load records
State, Zone, Disease, Cases, PrevCases
Group records by State
Total(State) = sum of Cases
FOR EACH record:
Percent = Cases / Total x 100 (1 dp)
Growth = (Cases - PrevCases) / PrevCases x 100
(Growth = 0 when PrevCases = 0)
Growth >= 50 AND
Percent >= 10 ?
YES
Alert = OUTBREAK
NO
Growth >= 25 ?
YES
Alert = WATCH
NO
Alert = NORMAL
Store result row
(State, Disease, Cases, Percent, Growth, Alert)
repeat until all records are processed
Aggregate selection: top disease, total cases,
number of OUTBREAK alerts
OUTPUT: statistics table, share chart,
trend chart, outbreak alert list, CSV export
END

9. Outputs

Output	Description
Disease statistics table	State, zone, disease, cases, percent, growth and alert for the selected state and disease
Share chart	Bar chart of each disease's percentage of reported cases
Trend chart	Line chart of disease cases over consecutive periods (the current script compares two periods; the mockup shows six weekly points as the proposed extension)
Outbreak alert panel	List of OUTBREAK and WATCH records with cases, share and growth, colour coded red and amber
Summary line	Most prominent disease, total cases and number of outbreak alerts
CSV export	The current selection saved as a report file

10. Assumptions and Limitations

- Percentages describe a disease's share of *reported* cases within a state, not infection rates per population.

- Growth depends on the quality and timeliness of reports; a very small previous value can give a large percentage.

- Alert thresholds are simple fixed values and are not a substitute for professional epidemiological judgement.

- The built-in data is sample data for demonstration. Real use needs official, regularly updated data (for example from NCDC).

11. GUI Mockups

Two mockup images accompany this document: **GUI_Mockup_1_Dashboard.png** (state dashboard with statistics, percentages, trend chart and alerts) and **GUI_Mockup_2_National_Overview.png** (all states with status and alerts).